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Towards a Smarter Charging Infrastructure: Real-Time Availability Forecasting for EVs

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Abstract

The electric vehicle (EV) market is experiencing substantial growth, and it is anticipated to play a major role as a replacement for fossil fuel-powered vehicles in transportation automation systems. Nevertheless, as a rule of thumb, EVs depend on electric charges, where appropriate usage, charging, and energy management are vital requirements. Examining the work done before gave us a reason and a basis for making a system that forecasts the real-time availability for EV charging stations that uses a scalable prediction engine built into a server-side software application that many people can use. The implementation process involved scraping data from various sources, creating datasets, and applying feature engineering to the data model. We then applied fundamental machine learning models to the pre-processed dataset, and subsequently, we implemented an ensemble model that combines the strengths of both Random Forest (RF) and Artificial Neural Network (ANN). This approach leverages the RF's resilience to overfitting and its ability to handle diverse data while benefiting from the ANN's capacity to capture complex non-linear relationships. The resulting ensemble model demonstrates significant improvements in precision, recall, and F1-score compared to individual models, making it a valuable contribution to the transportation sector.

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1. Introduction

Amidst growing concerns about environmental issues and fossil fuel pollution, the EV industry is experiencing significant global growth and resilience, as highlighted in the 2023 Global EV Outlook report [1]. Despite challenges posed by the COVID-19 pandemic, global EV sales surpassed 10 million units in 2022, even with a 16% decline in overall car sales. China leads the global electric car market with over 50% of all EV worldwide [2] and has success-

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fully achieved its 2025 new energy vehicle sales target, demonstrating its commitment to sustainable mobility and combating environmental challenges [3].

The growing prevalence of EV relies heavily on the charging infrastructure, known as charging stations or electric vehicle supply equipment (EVSE), which plays a critical role in facilitating the charging process for various plug-in EV [4]. Studies have shown that the availability and quality of charging stations strongly influence EV adoption [5, 6], and insufficient or low-quality stations are obstacles to ownership, even in countries with high home charging rates like Norway [7, 8, 9]. In Sweden, there is a correlation between the surge in public charging points, particularly in urban regions, and a rise in the adoption rates of EV [10]. A widespread and standardized charging infrastructure is essential for EV growth and acceptance [11, 12]. It ensures reliable access, promotes EV adoption, eliminates range anxiety, and offers convenience through interoperability across various locations and vehicle models.

Several studies have explored the application of machine learning and deep learning in various aspects of EV charging infrastructure. Soldan et al. [13] propose a data stream analysis approach for short-term prediction of EV charging station occupancy using logistic regression and real-time updates. Sao et al. [14] introduce the Deep Fusion of Dynamic and Static Information Models (DFDS) to improve forecasting accuracy by integrating static and dynamic information with a Gated Recurrent Unit (GRU). Hecht et al. [15] investigate Ensemble machine-learning techniques, specifically Gradient Boosting Decision Trees (GBDTs) and Random Forest classifiers, for predicting EV charging station accessibility, highlighting the importance of accurate historical data. Additionally, research by [16] proposes an LSTM neural network with feedforward layers to predict public charging station occupancy in Dundee City, UK, achieving enhanced accuracy by incorporating multiple data sources and daily statistics. These studies demonstrate the potential of machine learning and deep learning for optimizing EV charging infrastructure and improving the user experience.

In comparing these studies, Soldan et al.'s [13] data streaming approach offers real-time updates, which could be advantageous for dynamic charging environments. Still, their model's performance might differ with varying data quality and size. Sao et al.'s [14] DFDS model demonstrated the potential of combining static and dynamic information, but its application in diverse geographical regions remains to be tested. Hecht et al.'s [15] Ensemble approach showcased promising accuracy, but the effectiveness might vary with different datasets and charging station types. Finally, the LSTM model proposed by Ma et al. [16] showed promise in intraday forecasts, but its generalizability to more extensive and diverse datasets warrants further investigation. These studies provide valuable insights into EV charging station occupancy forecasting, but additional research is necessary to validate and refine their methodologies under various real-world scenarios.

Several studies have explored the use of machine learning and deep learning methodologies to analyze charging behaviors and predict future EV charging demand, complementing research on charging station availability. Qiao et al. [17] developed predictive models for forecasting charging demand, considering both registered and unregistered users, and found XGBoost to be the most effective method. Shahriar et al. [18] focused on predicting charging behavior, session duration, and energy consumption to address power grid strain, employing various machine learning algorithms and the ACN dataset [19]. Both studies provide valuable insights into charging behavior and demand prediction, with XGBoost demonstrating robustness in both cases. However, considering dataset specifics and use cases is crucial for assessing model applicability in diverse environments. Future research could integrate these insights to develop comprehensive models that address user behavior, charging patterns, and power grid challenges.

Several studies have significantly advanced research in EV charging demand forecasting. Almaghrebi et al. [20] developed a model using various machine learning methods and evaluated its performance on a real-world dataset. Kim et al. [21] compared different modeling techniques, incorporating historical data and external variables for demand prediction. Zhao et al. [22] proposed a data-driven framework to address overfitting issues in complex environments with limited data, comparing their model to high-performance counterparts using real-world EV data. Zhu et al. [23, 24] developed an LSTM-based deep learning algorithm for EV charging demand prediction, demonstrating its superiority over traditional time-series forecasting methods. These studies collectively contribute to a deeper understanding of EV charging demand and provide valuable insights for developing accurate forecasting models.

Various studies have explored EV charging demand prediction, each with its own strengths and limitations [2]. While Zhao et al.'s [22] data-driven framework addresses overfitting and enhances model reliability, it focuses on specific high-performance models, potentially overlooking other suitable techniques. Almaghrebi et al. [20] explore various machine learning methods but could benefit from further analysis of model interpretability and feature impor-

tance. Kim et al.'s [21] comparison of different techniques aids model selection, but a deeper exploration of model robustness in diverse environments is needed. Zhu et al.'s [23, 24] LSTM-based algorithm outperforms traditional methods, yet its focus on short-term predictions limits its practicality for long-term planning. While these studies contribute significantly to advancing EV charging demand prediction, enhancing model interpretability, evaluating robustness, and considering both short-term and long-term prediction horizons are crucial for improvement [2]. Additionally, addressing the research gap by focusing on predicting charging station availability and overcoming data limitations and scalability challenges is essential for supporting the growing EV ecosystem.

Current research on forecasting EV charging stations primarily utilizes statistical models [25, 21] and machine learning algorithms [26, 27, 28] to predict demand and optimize station placement, contributing significantly to sustainable transportation infrastructure development. However, a notable gap exists in accurately forecasting station availability, hindered by limited data, complex user behavior, and diverse charging patterns. This paper aims to address these limitations by focusing on the development of a system for accurate EV charging station availability prediction. Through web scraping data collection, the system utilizes machine learning and deep learning models to generate predictions, which are then integrated into a distributed software system providing real-time information to EV drivers. This approach tackles the challenges of accuracy and scalability, offering a valuable solution for enhancing the EV charging experience.

The rest of the paper is structured as follows: Section 2 presents our proposed methodology. The outcomes of our experimental evaluation are discussed in Section 3. Finally, Section 4 summarizes our takeaway messages and alludes to future issues.

2. Methodology

In this section, we introduce our methodology for predicting the status of EV charging stations using an ensemble model comprised of an ANN and an RF classifier. This ensemble approach aims to leverage both models' strengths, enhancing our prediction system's accuracy and robustness. The ANN model, implemented using Python's Keras deep learning library with a sequential architecture, excels at capturing complex non-linear relationships within the data. Its versatility as a universal function approximator makes it well-suited for handling diverse and intricate patterns present in charging station data. The RF model, on the other hand, is known for its robustness to overfitting, making it a strong complement to the ANN. We train both the ANN and RF models separately on a dataset sourced from a CSV file containing comprehensive information about various charging stations [2]. This dataset is meticulously preprocessed to prepare it for model training involving feature engineering and data transformations. The performance of each individual model and the ensemble as a whole is evaluated using several metrics, including F1-score, precision, and recall. Visualization techniques are employed to represent the prediction results in an easily interpretable manner. By combining the predictive capabilities of the ANN and RF, we aim to create a robust system that can reliably forecast the availability of EV charging stations. This ensemble approach leverages the strengths of each individual model, enhancing the overall accuracy and providing a more reliable prediction system for EV drivers.

Our primary goal is to create a robust system capable of reliably predicting the availability of charging stations specifically designed for EV. To achieve this objective, we employ web scraping techniques to gather relevant data, which is processed and transformed into machine learning and deep learning models. These models play a vital role in accurately forecasting the availability of EV charging spots. By leveraging the power of advanced computational algorithms, we can analyze historical data and identify patterns that contribute to predicting charging station availability with a high degree of accuracy. Combining data collection, machine learning, deep learning techniques, and real-time information dissemination integration, we aim to deliver a reliable and user-friendly system that empowers EV drivers with up-to-date knowledge about charging station availability, enhancing their overall charging experience. In the following, we will provide a detailed account of each step mentioned above, including data scraping, dataset generation, ML/DL models for prediction, and EV charging spot availability prediction.

The proposed model for EV charging station forecasting follows a structured architecture, as illustrated in Figure 1. The process begins with data scraping from various sources to gather comprehensive information. Next, feature engineering transforms raw data into meaningful features crucial for accurate predictions. The prepared features are then utilized in the model training stage, where machine learning algorithms like K-Nearest Neighbors, logistic regression, random forest, and support vector machines are implemented to develop robust predictive models. Once

trained, the model provides real-time predictions, informing EV drivers about charging station availability. This architecture emphasizes the interconnectedness of data scraping, feature engineering, and model training in achieving reliable predictions for EV charging station accessibility.

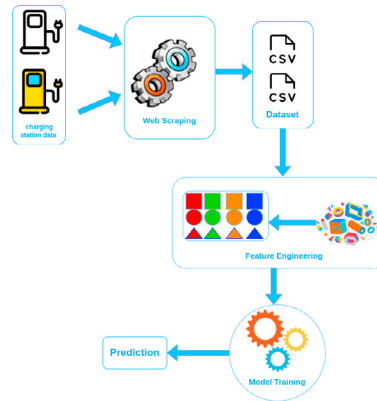


Fig. 1: The overall architecture of the proposed approach

To put the issue in simpler terms, Figure 2 illustrates how the model is designed to forecast the availability rate for charging stations along the driver's route and present this information to the driver. Notably, the third station stands out with the highest availability rate, making it the top recommendation for the driver.

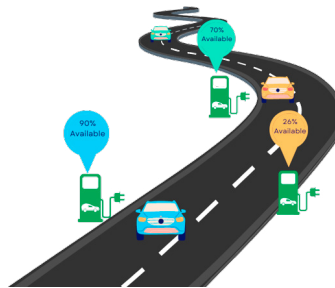


Fig. 2: Model simplification

The initial step of this study involved data scraping from two distinct sources to collect information on EV charging stations. The scraping process involved retrieving human-readable information from websites or programs and storing it in various formats [29]. The first dataset was sourced from OpenData Paris "Belib"[30], which offers a publicly accessible API [31]. This API facilitated structured data retrieval, allowing us to collect real-time geolocation data on 1,822 EV charging stations in Paris, France. Data was collected at 30-minute intervals and stored in CSV format. The resulting dataset comprised over 3,600,000 rows, each representing an individual charging station with specific details:

- "id _chargingStation": The ID of the charging station
- "status _chargingStation": The status of the charging station
- "station _address": The address of the charging station
- "station _postal _code": The postal code of the charging station location
- "station _latitude": The latitude coordinate of the charging station
- "station _longitude": The longitude coordinate of the charging station
- "last _update _time": The date and time when the data was last updated for the charging station

- "data_record_timestamp": The timestamp when the data was recorded

The second dataset was obtained from Enefit VOLT, a private Estonian company operating a network of EV charging stations [32]. Since Enefit VOLT does not provide a public API, we employed a more complex scraping approach "HTML Parsing and Asynchronous Request Imitation". This involved mimicking asynchronous requests and parsing the HTML structure of their website, yielding data on over 185 stations across Estonia. This approach was necessary because Enefit VOLT does not offer a publicly accessible API. The collected data for each charging station is then stored in a CSV file, resulting in over 370,000 rows of data, each representing a single EV charging point with the following attributes[2]:

- "id_chargingStation": Charging station ID
- "chargingStation_name": Charging station display name
- "status_chargingStation": Status of the charging station
- "station_address": Charging station address (address and city)
- "charge_rate": Charging speed of the station
- "maintenance_status": Indicates if the station is under maintenance
- "station_lat": Latitude coordinate of the station location
- "station_lon": Longitude coordinate of the station location
- "access_permission": Station access level
- "outlet_count": Number of outlets available at the station
- "outlet_1 and outlet_2": Information for two outlets (name, status, type, charging mode, maximum power, and kWh price)

The data collection timestamp is also recorded using `datetime.now()` for each entry. Once the data was scraped, each dataset underwent feature engineering [33] to prepare them for machine learning. This process involved transforming raw data into useful features for the model, including selecting the most pertinent predictor variables. Both the Belib and Enefit VOLT datasets shared common steps, such as extracting temporal features (year, month, day, hour, minute, second) from timestamps, removing redundant columns, and normalizing latitude and longitude values using MinMax scaling.

However, each dataset required specific feature engineering techniques. The Belib dataset utilized one-hot encoding for temporal features, converting categorical time information into a numerical representation. In contrast, Enefit VOLT employed label encoding for categorical features like charging speed and access level. Additionally, MinMax scaling was applied to numerical features like socket power and price in the Enefit VOLT dataset. Both datasets used one-hot encoding for temporal features. Finally, after feature engineering, the processed datasets were combined into a single CSV file. This unified dataset, now appropriately structured for machine learning, was split into training and testing sets to evaluate model performance on unseen data.

This comprehensive approach ensured both datasets were prepared for machine learning, enabling models to learn patterns and accurately predict charging station availability.

The core of our methodology is a powerful Ensemble model, specifically a combination of an RF and an ANN. This strategic pairing leverages the unique strengths of both models. The RF, recognized for its robustness, ability to handle diverse data types, and resistance to overfitting, excels at capturing complex patterns in the data. We employ a standard RF architecture with a specified number of decision trees and a defined maximum depth for each tree. This configuration allows for a balanced trade-off between model complexity and potential overfitting. The ANN, on the other hand, brings its flexibility and capability to learn non-linear relationships, particularly in real-time, to the table. Our ANN architecture includes a dense hidden layer with 256 neurons employing the Rectified Linear Unit (ReLU) activation function. ReLU introduces non-linearity, allowing the ANN to effectively handle complex patterns and extract meaningful features from the input data. The output layer, in turn, utilizes the softmax activation function, which provides probability distributions for each charging station status category, aiding in multiclass classification.

Once the model architecture is established, we train both the ANN and RF models on the prepared training dataset. The ANN is trained for 20 epochs with a batch size of 10, using the SGD optimizer with a learning rate of 0.3. The RF model is trained using default parameters. We then combine the predictions from both models by averaging their

predicted availability percentages, creating an ensemble approach that leverages their strengths and potentially leads to more robust and accurate predictions. Finally, we evaluate the performance of this ensemble model on the testing dataset.

In conclusion, our methodology centers on an ensemble model integrating an ANN and RF classifier to accurately predict EV charging stations' status. This approach combines the ANN's ability to capture non-linear patterns with the RF's robustness and versatility, resulting in a more reliable and accurate prediction system.

3. Results and Discussion

This section presents the results of evaluating the performance of multi-class classification machine learning and deep learning models. The performance of these models is assessed using precision, recall, and F1 score metrics to measure and compare their effectiveness in predicting charging station availability.

Table 1 displays the outcomes obtained through the machine learning/deep learning methodology applied to the Paris dataset. The results demonstrate that among the ML baseline models, RF achieved the highest performance across all metrics, with an accuracy rate exceeding 95%. K-Nearest Neighbors (KNN) followed closely with a rate of 94% in all metrics. Regarding the F1 Score, the Support Vector Machine (SVM) yielded the lowest rate at 59.54%, with Logistic Regression closely behind at 60.73%.

However, the ANN model, despite slightly trailing behind the KNN and RF models in performance scores, managed to attain a precision of 89.46%, a recall of 87.94%, and an F1 score of 88.69%. It's important to note that the Combined (ANN+RF) model achieved a competitive performance, reaching a recall of 91.21%, a precision of 92.87%, and an F1-score of 92.21%, demonstrating its ability to leverage the strengths of both the RF and ANN models.

Based on the results presented in Table 1, the RF and KNN models are the top performers on this dataset, as they consistently achieve high precision, recall, and F1 scores. The Logistic Regression and SVM models show weaker performance. At the same time, the ANN model also performs well, falling in between the top-performing models and the weaker ones. While it's noteworthy that ANN forecasts rates for all potential outcomes and offers real-time predictions, the instance-based learning of KNN constrains its ability to provide real-time predictions. Additionally, RF doesn't furnish rates for output classes, presenting results solely. The combined (RF+ANN) model, however, demonstrates the potential of ensemble methods to enhance both accuracy and the balance between precision and recall. This suggests that ensemble approaches may be particularly valuable in scenarios where individual models exhibit strengths and weaknesses, as the ensemble can potentially mitigate those weaknesses while capitalizing on individual model strengths.

Table 1: Performance results (%) per model for the Belib Dataset

APPROACH	RECALL	PRECISION	F1-SCORE
ARTIFICIAL NEURAL NETWORK	87.94	89.46	88.69
K-NEAREST NEIGHBOURS	94.13	94.10	94.11
LOGISTIC REGRESSION	72.16	66.65	60.44
RANDOM FOREST	95.56	95.54	95.41
SUPPORT VECTOR MACHINE	71.90	66.41	59.54
COMBINED (ANN+RF)	91.21	92.87	92.21

The figure 3 shows the performance of three different machine learning algorithms (ANN, RF, and the combined model) across five predictions using the Paris dataset. The ANN model starts with a high accuracy of around 94% and then drops significantly to 65% for the 4th prediction before recovering to 100% for the 5th prediction. The RF model starts at a slightly higher accuracy of 96% but then falls to 92% and 81% for the 2nd and 4th predictions, respectively. The RF model then reaches its highest performance of 98% for the 5th prediction. The combined model consistently performs better than the individual models, with its lowest accuracy (73%) being for the 4th prediction. The combined model also shows a good performance for the 5th prediction (99%). In general, the combined model outperforms the individual models.

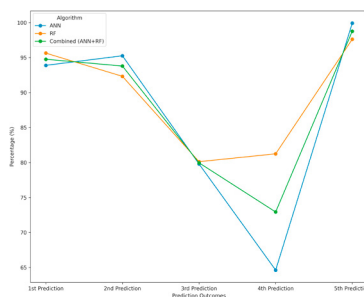


Fig. 3: Prediction Outcomes Comparison for belib

The findings from applying machine learning and deep learning approaches to Estonian data are presented in Table 2. The KNN model performs consistently across all metrics. It achieves high precision, recall, and F1-score, indicating its strong ability to correctly classify instances of different classes. The Logistic Regression model shows competitive performance. It achieves high recall, indicating its effectiveness in identifying positive cases, and maintains a balanced F1 score. The RF model demonstrates solid performance across all metrics. It achieves high recall and precision, resulting in a strong F1 score. The SVM model performs well, with high precision and recall values. The F1 score also reflects a good balance between the two metrics. The ANN model achieves impressive results, particularly in precision. Its recall and F1-score are also high, indicating its effectiveness in classification tasks on this dataset. To further investigate the potential for performance enhancement, an ensemble model combining the RF and ANN models was implemented. This combined model yielded a recall of 95.96%, a precision of 95.59%, and an F1-score of 95.77%, showcasing its ability to leverage the strengths of both individual models and maintain a high level of overall performance.

Based on the results presented in Table 2, all models exhibit strong performance on the "Enefit Volt Dataset". The ANN, Logistic Regression, and SVM models consistently achieve high precision, recall, and F1 scores. The KNN and RF models also perform well but have slightly lower F1 scores than the others. Notably, the combined (RF+ANN) model achieves competitive performance with an F1-score of 95.77%, demonstrating ensemble methods' potential to enhance prediction accuracy further, even when individual models already exhibit strong performance. Overall, these models, including the combined model, appear to be well-suited for classifying instances in the "Enefit Volt Dataset", indicating the effectiveness of machine learning approaches for this task.

Table 2: Enefit Volt Dataset Performance Results (%)

APPROACH	RECALL	PRECISION	F1-SCORE
ARTIFICIAL NEURAL NETWORK	95.74	96.04	95.89
K-NEAREST NEIGHBOURS	94.19	94.11	94.15
LOGISTIC REGRESSION	96.63	94.35	94.81
RANDOM FOREST	96.02	94.78	95.17
SUPPORT VECTOR MACHINE	96.32	95.15	95.75
COMBINED (ANN+RF)	95.96	95.59	95.77

The figure 4 shows the performance of three different machine learning algorithms (ANN, RF, and the combined model) across five predictions. The ANN model has a highly fluctuating performance. While starting with a high accuracy of around 99% for the first prediction, it drops to 70% for the 2nd prediction and then to 47% for the 3rd. However, it shows a remarkable recovery, reaching near 100% for the 4th and 5th predictions. The RF model has relatively stable performance across the predictions, starting at 89% for the 1st, slightly decreasing for the 2nd, and then increasing to 73% for the 3rd. It then shows a high accuracy of 99% for the 4th and finally 98% for the 5th prediction. The combined model also has a very high performance across the predictions. It consistently outperforms

both the individual models, with its lowest accuracy (69%) being for the 2nd prediction. Overall, the combined model shows a better and more stable performance than the individual models.

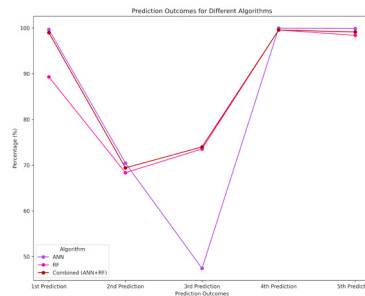


Fig. 4: Prediction Outcomes Comparison for Enefit Volt

The results of this paper demonstrate that the precision, recall, and F1 scores for all evaluation metrics surpass those of existing works in the literature review. These findings are particularly noteworthy given the differences in dataset features, which account for the varying evaluation scores of the models for the Paris and Estonia datasets. Notably, the Estonia dataset is deemed more robust due to its inclusion of additional features such as price per kWh and outlet types, contributing to its higher overall scores than the Paris dataset.

While evaluating the models on the Paris dataset, RF and KNN achieved slightly higher scores. However, several models offer distinct advantages depending on project priorities. The ANN excels in providing real-time predictions, making it ideal for time-sensitive applications. Additionally, ANN presents results as percentages for each category, offering detailed classification probabilities. This makes it a strong choice for tasks requiring a deeper understanding of outcomes. It's important to note that the Combined (ANN+RF) model emerged as a strong contender, potentially combining the strengths of both real-time predictions and informative classifications. Ultimately, the best model selection hinges on specific needs. ANN or the Combined (ANN+RF) model might be ideal for real-time needs and detailed classifications. On the other hand, if interpretability is crucial and real-time predictions are less important, simpler models like RF could be preferred. Regardless of the choice, all evaluated models demonstrate promise for various classification problems. Future research with even richer datasets can potentially improve further performance, particularly for models like ANN and the Combined (ANN+RF) that can effectively leverage complex features.

4. Conclusion

This paper's initial focus was conducting a thorough literature review and analyzing state-of-the-art research on EV charging station availability prediction, including its strengths, weaknesses, models, methods, and datasets. A comprehensive literature review has established a foundation and rationale to construct a scalable prediction engine for real-time availability forecasting in EV charging stations. The implementation process involved scraping data from various sources, creating datasets, and applying feature engineering to the data model. We then implemented an ensemble model combining the strengths of the RF and the ANN models. This ensemble approach leverages the complementary strengths of both models: the RF's resilience to overfitting and its ability to handle diverse data types, and the ANN's capacity to capture complex non-linear relationships. This combination results in a prediction engine that delivers even more accurate and reliable EV charging station availability estimates. The performance of the ensemble model was evaluated using standard metrics, and the results demonstrate significant improvements compared to using individual models alone.

The accuracy of our EV charging station availability prediction system relies heavily on the quality and availability of data. Inconsistent, outdated, or missing data can lead to inaccurate predictions, especially in real-time scenarios. While our ensemble approach helps mitigate overfitting, there is still a risk that the models might be overly tailored to the specific training datasets, potentially hindering their ability to generalize well to new or different EV charging environments. Addressing these limitations through robust data collection methods and further optimization of the

ensemble model are key priorities for future research. There are several potential avenues for future work to enhance the work presented in this paper while also addressing the challenges posed by controlling the uncertainty of the dynamic arrival of EV charging requests.

- The manual nature of the data scraping and feature engineering components suggests that automation could yield further improvements. Specifically, it may be possible to automatically identify which features are most important and which are less critical and to log differences that arise when a particular feature is excluded. Engaging in this process additionally improves the quality, precision, and dependability of the model.
- This study will also be extended to a larger scope, which is a better understanding of capacity fade, i.e., the battery's ability to hold a charge diminishes over time. Beyond intrinsic features like electrode degradation and solid-electrolyte interface growth, we would also be eager to explore the driving quality impact further.
- Future work will investigate the potential of incorporating other machine learning models into the ensemble, such as Gradient Boosting Machines or Support Vector Machines, to explore a wider range of predictive capabilities.
- Last but not least, while the prediction engine is currently limited to Paris and Estonia, it could be readily adapted to other cities or countries.

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